

# Hadoop Ecosystem and Yarn

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# 1. Hadoop Ecosystem Components

The Hadoop Ecosystem is a collection of open-source projects that work together to provide a complete Big Data solution. The major components are:

## A. Core Components

- HDFS (Hadoop Distributed File System)
- MapReduce
- YARN (Yet Another Resource Negotiator) management

| Component | Function                                                                                                  |
|-----------|-----------------------------------------------------------------------------------------------------------|
| Hive      | Data warehouse infrastructure; converts SQL-like queries into MapReduce jobs.                             |
| Pig       | A high-level scripting language for creating complex data transformations that can be executed on Hadoop. |
| HBase     | NoSQL database built on top of HDFS; column-oriented and supports random read/write.                      |
| Sqoop     | Transfers data between Hadoop and relational databases (import/export).                                   |
| Flume     | Collects, aggregates, and moves large amounts of log data into HDFS.                                      |
| Oozie     | Workflow scheduler for Hadoop jobs.                                                                       |
| Zookeeper | Distributed coordination service.                                                                         |
| Mahout    | Machine learning library on Hadoop.                                                                       |
| Ambari    | Cluster management tool for monitoring, and managing Hadoop clusters.                                     |

**Table:** Supporting Components in Hadoop Ecosystem

## 2. Hadoop Schedulers

### Types of **Hadoop** Schedulers



Figure: Hadoop scheduler

# Hadoop 2.0 – New Features

Hadoop 2.x was a major upgrade over Hadoop 1.x, solving scalability, fault-tolerance, and flexibility issues. The key improvements include:

- NameNode High Availability
- HDFS Federation
- MapReduce v2 (MRv2)
- YARN

# NameNode High Availability (HA):

- Introduced Active/Standby NameNodes to eliminate SPOF(Single Point of Failure).
- Two NameNodes run in the cluster:
  1. Active NameNode – Handles all client requests.
  2. Standby NameNode – Maintains a synchronized copy of metadata and takes over if Active fails.

# HDFS Federation :

Introduced to address the memory limitations of a single NameNode, HDFS Federation allows for the horizontal scaling of the NameNode service.

- Federation allows multiple independent NameNodes to manage different parts of the namespace.
- Each NameNode has its own Namespace Volume.
- DataNodes register with all NameNodes, but store blocks for multiple namespaces.

# MapReduce v2 (MRv2) :

Separates resource management from job execution.

JobTracker is split into:

- ResourceManager (RM)– Allocates resources across the cluster.
- ApplicationMaster (AM)– Manages job execution for a single job.



# YARN (Yet Another Resource Negotiator)

YARN is the biggest innovation in Hadoop 2.x and serves as the cluster operating system.

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- Each NameNode has its own Namespace Volume.
- DataNodes register with all NameNodes, but store blocks for multiple namespaces.

| Criteria                    | Hadoop 1.x                                        | Hadoop 2.x                                                               |
|-----------------------------|---------------------------------------------------|--------------------------------------------------------------------------|
| Architecture                | Master-slave                                      | Multi-node cluster setup                                                 |
| Resource management         | JobTracker for resource management and scheduling | Yet Another Resource Negotiator (YARN) for resource management           |
| Number of NameNodes         | Single NameNode                                   | Multiple NameNodes (high availability)                                   |
| Scalability                 | Supports up to 4,000 nodes                        | Supports more than 10,000 nodes                                          |
| Fault tolerance             | Limited to secondary NameNode                     | NameNode high availability with automatic failover                       |
| Job scheduling and tracking | JobTracker manages both                           | Resource Manager for scheduling and Application Master for tracking      |
| MapReduce                   | Tightly integrated                                | Just one of the many applications under YARN                             |
| Data processing model       | Only MapReduce                                    | Supports various data processing models like graph processing and others |
| Cluster utilisation         | Underutilised in some cases                       | Better utilisation due to multiple workloads and generic resource model  |

Figure: Difference between Hadoop 1.x and Hadoop 2.x